EngineWatch
AEROSPACE INTELLIGENCE

INTRODUCING

# EngineWatch

An **AI-powered** aerospace predictive maintenance platform.

- **Aerospace Focus**
- **Predictive Maintenance**
- **Estimate Remaining Engine Life (RUL)**

*Turning engine telemetry into early, actionable maintenance insights.*



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⚠ THE PROBLEM

# WHY ENGINE MAINTENANCE MATTERS?



Aircraft engines are critical, high-value systems. Timely maintenance keeps them **safe, reliable**, and **operationally efficient**.

## Limitations of Fixed-Interval Maintenance

- Over-maintenance (higher cost)
- Misses accelerated wear
- Ignores real-time engine condition

## Why Predict Degradation?

- Uses actual sensor telemetry
- Catches early signs of wear
- Enables safer, smarter, and cost-efficient maintenance



*Move from a fixed schedule to a condition-based, predictive approach.*

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THE SOLUTION



# HOW ENGINEWATCH WORKS

A data-driven ML pipeline that turns engine sensor telemetry into reliable **Remaining Useful Life (RUL)** predictions.

## NASA C-MAPSS Dataset

Turbofan engine degradation simulation (FD001)

## Sensor Data & Feature Engineering

- 14 engine sensor readings
- Rolling statistics (trend & drift)
- Engine-wise data preparation
- Capped RUL target

## Random Forest Machine Learning

- Ensemble of decision trees
- Captures non-linear degradation
- More accurate & robust than linear models

## RUL Prediction via FastAPI

FastAPI

- Real-time inference endpoint
- Input validation (Pydantic)
- Returns predicted RUL and engine health tier (**Healthy** / **Watch** / **Critical**)

Engine Telemetry → Data Processing → Trained ML Model → FastAPI Inference → **RUL Estimate + Health Status**

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TECHNOLOGY & CAPABILITIES

# TECHNOLOGY STACK & PROJECT CAPABILITIES

A modern, full-stack AI platform that combines machine learning, real-time APIs, and **interactive aerospace visualization.**



### Frontend (Web Platform)
Next.js 14, TypeScript, React, Tailwind CSS

### Backend (API & Services)
Python, FastAPI, Pydantic, Uvicorn

### Machine Learning
Scikit-learn, Random Forest, NumPy, Pandas

## KEY CAPABILITIES

**Interactive Operations Dashboard**
Real-time engine health and RUL insights

**3D Turbofan Digital Twin**
Interactive engine visualization & components

**Degradation & Sensor Analytics**
Multi-sensor trends and performance drift

**Fleet & Flight Operations View**
Live fleet monitoring and airspace visualization

RUL Prediction | Real-time Inference | Interactive Visualization | Scalable & Secure